

Muhammet Yunus Seker

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Education

- 2022–Current **Carnegie Mellon University, The Robotics Institute, PhD, USA, GPA: 4.00/4.00**
- 2017–2020 **Bogazici University, Computer Engineering, MSc, Turkey, GPA: 3.7/4.00**
- 2012–2017 **Bogazici University, Computer Engineering, BSc, Turkey, GPA: 3.42/4.00**

Skills

- Technical Python, C++, Java, ROS, PyTorch, Keras, Tensorflow, Isaac Lab, CoppeliaSim, Gazebo, Mujoco
- Robots UR5e, GP4 Yaskawa, Franka, UR10, Baxter, Nao, Lego Mindstorms

Research Experience

- 2022–Current **Research Assistant, Intelligent Autonomous Manipulation Lab, Carnegie Mellon University, USA, Supervisor: Prof. Oliver Kroemer**
- Currently developing methods for zero-shot skill transfer in manipulation tasks by integrating modern foundational models and vision-language models (VLMs/LLMs) with classical task-axis controllers
 - The objective is to enable zero-shot transfer of task-axis controllers across diverse environments by leveraging AI-grounded keypoints and axes, allowing semantic generalization of skills to relevant objects and parts
 - Utilizing visual foundation models such as DINOv3, SAM, Gemini-ER, and other VLM/LLMs to enhance perception, scene understanding, skill planning, and task adaptability
 - Previously contributed to a collaborative project with Sony AI on precise food plating and manipulation, which involved optimizing material properties of deformable simulation objects using Bayesian Optimization and executing precise plating through multi-modal planning
- 2017–2022 **Research Assistant, Cognition, Learning and Robotics (COLORS) Lab, Bogazici University, Turkey, Co-supervisors: Prof. Justus Piater, Prof. Emre Ugur**
- Focused on developing and improving advanced methods for robot learning using deep neural networks with supervised and self-supervised learning
 - Conducted research on imitation learning (learning from demonstration), RL, affordance learning, action-effect prediction, multi-modal systems, representation learning, and symbol emergence
 - Contributed to the EU H2020 IMAGINE Project, aiming to enable robots to understand environmental structure and the effects of their actions
 - Designed an association engine and developed algorithms for predicting object affordances, action parameters, and action effects using deep learning and vision-based models for electronic device disassembly
 - Supervised three undergraduate students on their research projects
- 2020 Jan–Mar **Research Intern, Cognitive Developmental Robotics Lab (Nagai Lab), University of Tokyo, Japan, Co-Advisors: Prof. Yukie Nagai, Prof. Erhan Oztop**
- Studied imitation and mirror neuron systems using deep learning and conditional neural processes
 - Proposed a neural network structure that allows simultaneous learning of multi-modal representations

Teaching Experience

- Summer 2025 **Teaching Assistant, CS 16-720 Computer Vision, Carnegie Mellon University, USA**
- Worked with Prof. Deva Ramanan to lecture 96 students, contributing to assignment grading, and office hours
- Fall 2024 **Teaching Assistant, CS 16-824 Visual Learning and Recognition, Carnegie Mellon University, USA**
- Supported Prof. Jun-Yan Zhu in managing coursework and evaluating student submissions for a class of 74
- 2021–2022 **Teaching Assistant, CMPE 150 Introduction to Computing, Bogazici University, Turkey**
- Designed and graded weekly programming tasks; assisted students with core coding concepts

Work Experience

- 2020–2022 **Spiky.AI, Machine Learning and AI Team Lead, Research Engineer, USA**
- Led the Machine Learning and AI team for two years, overseeing the development of advanced models for video and audio sentiment analysis
 - Directed projects involving face detection, face recognition, emotion recognition, gaze estimation, and speech-to-text processing
 - Applied a diverse range of techniques, including Transformer-based architectures, transfer learning, unsupervised face feature extraction and clustering, as well as Bayesian machine learning methods

Publications

- Humanoids 2025 **M.Y. Seker**, S. Aggarwal, O. Kroemer. "Grounded Task Axes: Zero-Shot Semantic Skill Generalization via Task-Axis Controllers and Visual Foundation Models", Humanoids, 2024
- IROS 2024 **M.Y. Seker**, O. Kroemer. "Leveraging Simulation-Based Model Preconditions for Fast Action Parameter Optimization with Multiple Models", IROS, 2024
- ICRA 2024 **M.Y. Seker**, O. Kroemer. "Estimating Material Properties of Interacting Objects Using Sum-GP-UCB" International Conference on Robotics and Automation, ICRA, 2024
- JAIR 2022 A. Ahmetoglu, **M.Y. Seker**, J. Piater, E. Oztop, E. Ugur. "Deepsym: Deep symbol generation and rule learning for planning from unsupervised robot interaction" Vol. 75, pp. 709-745, JAIR, 2022
- Neural Networks 2021 **M.Y. Seker**, A. Ahmetoglu, Y. Nagai, M. Asada, E. Oztop, E. Ugur. "Imitation and mirror systems in robots through Deep Modality Blending Networks" Vol. 146, ISSN 0893-6080, Neural Networks, 2021
- ICRA 2020 **M.Y. Seker**, E. Oztop, M.T. Akbulut, Y. Nagai, M. Asada, E. Ugur. "Towards a Mirror Neuron System via Dual Channel Conditional Neural Movement Primitives" ICRA Brain-PIL Workshop, 2020
- ICRA 2020 M. Kurtoglu, **M.Y. Seker**, E. Samur, E. Ugur. "Predicting Whole Body Motion Trajectories using Conditional Neural Movement Primitives" ICRA Long-term Human Motion Prediction Workshop, 2020
- CoRL 2020 M. Akbulut, E. Oztop, **M.Y. Seker**, X. Hh, A. Tekden, E. Ugur. "ACNMP: Skill transfer and task extrapolation through learning from demonstration and reinforcement learning via representation sharing" Conference on Robot Learning, pp. 1896-1907. PMLR, 2020
- ICRA 2020 A. Tekden, A. Erdem, E. Erdem, M. Imre, **M.Y. Seker**, E. Ugur. "Belief regulated dual propagation nets for learning action effects on groups of articulated objects." International Conference on Robotics and Automation, pp. 10556-10562, IEEE, 2020
- RSS 2019 **M.Y. Seker**, M. Imre, J. Piater, E. Ugur. "Conditional Neural Movement Primitives" Robotics: Science and Systems, vol. 10, 2019
- RAS 2019 **M.Y. Seker**, A. Tekden, E. Ugur. "Deep Effect Trajectory Prediction in Robot Manipulation" Robotics and Autonomous Systems 119, 173-84, 2019

Academic Activities

- 2019-2025 **Reviewer**,
Conference on Robot Learning, CoRL 2024-2025
International Conference on Robotics and Automation, ICRA 2019-2025
IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2019-2025
IEEE Robotics and Automation Letters, RA-L 2020-2025
Robotics: Science and Systems, RSS 2020
Transactions on Cognitive and Developmental Systems, TCDS 2020
International Conference on Advanced Robotics, ICAR 2020
- 2019-2025 **Oral and Poster Presentations**,
Humanoids 2025
CMU Robot Learning Workshop 2025
IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2024
International Conference on Robotics and Automation, ICRA 2024
Conference on Robot Learning, CoRL 2020
Brain-PIL Workshop on Brain-inspired Perception, Interaction and Learning, ICRA 2020
Robotics: Science and Systems, RSS 2019

Projects

- 2023 **Understanding Human Preferences on Verbal and Non-Verbal Modalities in HRI**
◦ Investigated enhancing communication clarity in Human-Robot Interaction (HRI)
◦ Explored three primary modalities: verbal clarification, user-centric pointing, and robot-centric pointing actions
◦ Employed human-involved experiments and surveys to analyze human preferences and feedback
- 2023 **Non-prehensile Manipulation for Shelf Organization, Carnegie Mellon University**
◦ Investigated non-prehensile robot pushing motions for re-organizing boxes on a shelf to optimize space
◦ Developed a classifier with depth images for assessing box placement feasibility
◦ Employed a search-based approach for planning robot trajectories and implemented it in a physics simulator
- 2017 **Project Avatar, Teleoperating NAO Robot**
◦ **Winner Project** for Bogazici University 2017 Graduation Project Competition
◦ Aimed to tele-operate a NAO robot in real time using Kinect motion sensor and VR glasses
◦ Investigated the body schema adaptation in humans in response to changes in their embodiment